Deokhyun **Ahn**

Ph.D. Candidate · Brain & Cognitive Engineering

Expected graduation: August 2027
Available for employment from March 2027

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01 Research Interests

Reliable and efficient video understanding for social good.

Research for social good has always been my first priority, and it will continue to guide my work. I am a Ph.D. candidate in Brain & Cognitive Engineering at Korea University and a student researcher at KIST AI·Robotics.

My early work on sensing applications, including an ultrasonic SLAM helmet for firefighters and indoor drone tracking, shaped my commitment to dependable systems for real-world needs. I now study reliable and efficient video understanding: identifying anomalous behavior, understanding its temporal context, and explaining it with visual evidence.

My work connects video anomaly detection and understanding, adaptive video sampling, action recognition, and vision-language models (VLMs). Through public-safety research at KIST, I work on contextual anomaly detection and behavior analysis for real-world surveillance video.

Recent work includes BUSTER for VLM-guided anomaly detection with adaptive sampling, Where and What for contextual anomaly detection, and Look Closer for action-guided proficiency estimation.

Current focus: why vision-language models hallucinate in video anomaly understanding, and how selective visual verification can make their explanations more faithful.

Video understanding | Video anomaly detection & understanding | Adaptive video sampling | Efficient vision-language models | VLM hallucination & reliability | Temporal & contextual reasoning | Action recognition & human motion analysis | Video surveillance & public safety

02 Education

Mar 2023 – Present
Seoul, KR

Ph.D. Candidate, Brain & Cognitive Engineering
Korea University

- Expected graduation: August 2027.
- Study contextual video understanding inspired by human cognitive processes.

Aug 2015 – Feb 2020
Seoul, KR

M.Eng., Electronic Engineering
Kwangwoon University

- Thesis: *A Study on Passive Target Tracking in Three-Dimensional Indoor Environment by Using Deep Neural Network*.
- Completed 19 months of mandatory military service.

Mar 2013 – Aug 2015
Seoul, KR

B.Eng., Electronic Engineering TRANSFER
Kwangwoon University

- Transferred in March 2013 from the National Institute for Lifelong Education after completing B.Eng. coursework in Computer Engineering from March 2011 to February 2013.

03 Research & Work Experience

Jan 2023 – Present
Seoul, KR

Student Researcher CURRENT

KIST AI·Robotics Institute · Vision Intelligence Group

Video anomaly understanding · Surveillance video · Vision language models

- Develop public safety video systems spanning action recognition, contextual anomaly detection, and vision language models that explain anomalous behavior. [1] [2] [3] [4] [5]
- Developed research prototypes and integrated them into a real-world monitoring system.
- Served as frontend team lead for the monitoring system, contributing directly to implementation and coordinating the team's development work.

Jan 2022 – Feb 2023
Seoul, KR

Research Intern

KIST AI·Robotics Institute · Vision Intelligence Group

Began surveillance video research through urban public safety projects and national research programs.

Oct 2021 – Dec 2021
Sejong, KR

Researcher

The Korea Transport Institute

Contributed to autonomous bus development through SLAM, control center systems, policy research, and coordination.

04 Research Grants & Projects

~$24M
Five year program

Real-Time Intelligent Surveillance for Public Safety NATIONAL R&D

MSIT & MOTIE · with Yonsei University and the Korean National Police Agency

Video anomaly detection that uses context across city scale CCTV control centers with about 1,000 cameras, validated in Anyang City.

~$9.5M
Five year program

Mobile Platform for Active Safety Services

Korean National Police Agency & MOTIE

Government research program using mobile platforms for proactive public safety services.

New
initiative

Anomaly Detection in Correctional Facilities

KIST AI·Robotics

Detecting inmate anomalous behavior in custodial environments.

05 Selected Publications

[1] **Ahn D.**, Kim H., Han J.-H., Ham B., Choi H., Kim I.-J., Kim H. — ACCEPTED
Look Closer: Action-Guided Dense Visual Dynamics for Proficiency Estimation.
CVPRW 2026 · Multimodal Human Motion Analysis · Pitch and Poster Presentation

[2] **Ahn D.**, Choi Y., Han J.-H., Kim I.-J., Kim H. — ACCEPTED
BUSTER: Adaptive Sampling for VLM-Guided Unsupervised Video Anomaly Detection.
British Machine Vision Conference (BMVC), 2026

[3] **Ahn D.**, Jo Y., Lee U.-S., Park H., Han J.-H., Kim H. — ACCEPTED
Learning Robust Representations for Few-Shot Action Recognition with Frame-Level Ambiguities.
IEEE Access, 2026

[4] **Ahn D.**, Jo Y., Kim D., Nam G. P., Han J.-H., Kim H. — PUBLISHED
Where and What: Contextual Dynamics-Aware Anomaly Detection in Surveillance Videos.
IEEE Transactions on Image Processing, 2025 · JIF 15.3 · Top 2.17% in E.E. Eng. (2025 JCR)

[5] Kim D., **Ahn D.**, Jo Y., Park H., Lee S., Kim H. PUBLISHED

A Foundational Research Framework for Real-World Abandoned Object Detection: Train-Free Baseline and a Standardized Benchmark.

Expert Systems with Applications, 2025 · 130658

[6] Zhou B., **Ahn D.**, Lee J., Sun C., Ahmed S., Kim Y. PUBLISHED

A Passive Tracking System Based on Geometric Constraints in Adaptive Wireless Sensor Networks.

Sensors, 18(10), 3276, 2018

[7] **Ahn D.**, Kim N. M., Park J. H., Kim Y. O. PUBLISHED

2D Indoor Map Building Scheme Using Ultrasonic Module.

J. Korean Inst. Comm. and Info. Sci., 41(8), 986–994, 2016

06 Awards & Honors

Dec 2025

AIR Outstanding Research Award

KIST AI·Robotics Institute

> In recognition of outstanding research achievements during 2025.

Feb 2025

Excellent Paper Award, Graduate School

Korea University

> For outstanding research performance and publication of an excellent paper during doctoral studies.

Nov 2024

Outstanding Oral Presentation Award

KIST Academia and Research Convergence Conference

> For the presentation “Multi-Aspect Contextual Dynamics for Anomaly Detection in Surveillance Videos.”

Jul 2021

Division Commander's Commendation

Capital Mechanized Infantry Division

> For a combat development proposal on reconnaissance drone AI improvement and acoustic detection using AI and big data.

07 Teaching

Sep – Dec 2018

Senior Capstone Project

Kwangwoon University · Instructor: Prof. Youngok Kim

- Assisted in managing and supporting the Capstone Design Project as a TA for 50+ students.

Mar – Jun 2017

Computer Architecture

Kwangwoon University · Instructor: Prof. Youngok Kim

- Graded assignments and provided feedback for 30+ students with Q&A sessions.

08 Technical Skills

Programming	Python C Java Git Linux
Machine Learning	PyTorch TensorFlow Keras NumPy Pandas Matplotlib
Sensors	Ultrasonic Beacon RF Wi-Fi 3-axis Accelerometer
Embedded & IoT	Raspberry Pi Arduino